

BRANCO ANTONIO MOLASSI

Avellaneda – Santa Fe, Argentina | brancomolassi17@gmail.com

Professional Profile

Electronic Engineer specialized in industrial automation and embedded systems development. 6 years of experience in designing and commissioning control solutions using **Siemens and Rockwell PLC/HMI** in high-demand production environments, complemented by embedded systems development based on **ESP32** and **STM32** with RTOS.

Featured Projects

Industrial Automation Systems – Meat Processing Plants (LatAm)

Responsible for the development and commissioning of control systems for high-capacity food processing plants.

- *Dourados – Brazil*
Production: 500 pigs/hour (≈ 9,000 pigs per day).
- *Asunción – Paraguay*
Production: 100 animals/hour (≈ 1,200 animals per day).
- *La Plata – Argentina*
Production: 200 animals/hour (≈ 3,000 animals per day).

Tecnologies: Siemens PLC & HMI, SCADA systems, industrial equipment integration via PROFINET and MODBUS.

Automatic Hose Cutting System (ESP32 + FreeRTOS)

- Full embedded solution development (hardware + firmware).
- Implementation of FreeRTOS-based architecture.
- DC motor control using PWM and encoder feedback.
- HMI interface on TFT display (ILI9488).

Professional Experience

Application Engineer

ASUAN SA (Frontmatec Group) | November 2024 – Present.

- Development and commissioning of PLC-based automation solutions for industrial applications.
- Design of HMI interfaces and SCADA systems focused on operation and diagnostics.
- Integration of equipment and control systems in production environments.
- Responsible for on-site commissioning and system validation.

Embedded Systems Developer (Freelance)

November 2025 – Present

- Design and development of custom embedded systems based on microcontrollers (ESP32, STM32).
- Firmware development using modular architecture and RTOS.
- Development of HMI interfaces and actuator control (DC motors, valves, sensors).
- Integration of industrial communication protocols (Modbus RTU/TCP).

Control Systems Developer

COLVEN SA | September 2023 – October 2024

- Development of control systems using PLCs and microcontrollers for industrial applications.
- Definition and validation of control systems.
- Approval of finished and semi-finished products.
- Innovation and improvement of control systems.

Industrial Programmer

Proyección Electroluz SRL | July 2020 – August 2023

- Development of wired and fiber optic industrial networks using MODBUS TCP/RTU, PROFIBUS, and PROFINET.
- PLC programming in industrial environments (SIEMENS, SCHNEIDER).
- Project follow-up and commissioning.

Technical Skills

- **Languages:** C / C++
- **Embedded Systems:** STM32, ESP32 (ESP-IDF, FreeRTOS)
- **Development Tools:** Visual Studio Code, Keil uVision
- **Design & Simulation:** Altium Designer, LTspice
- **Industrial Automation:** TIA Portal, Studio 5000
- **Documentation:** LaTeX

Education

- **Electronic Engineering (2014 – 2022)** National University of Córdoba (UNC)
- **Electromechanical Technician (2008 – 2013)** Technical School No. 451

Languages

- Spanish (Native).
- Portuguese (Advanced).
- English (Intermediate).